Interpretable by Design, Efficient by Distillation: An Industrial Study of Expert-Guided Search-Based Software Team Formation [Supplementary Material]

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I. Introduction

This document provides supplementary material for the paper *Interpretable by Design, Efficient by Distillation: An Industrial Study of Expert-Guided Search-Based Software Team Formation*. It preserves detailed experimental evidence summarized in the main paper without altering or recalculating any data.

Table S1 characterizes the industrial dataset used to calibrate and evaluate the pipeline, including the candidate pool, technical taxonomy, temporal project split, team-size distribution, and collaboration graph. Table S2 documents the retained practitioner-validation design and distinguishes unique cases, repeated comparisons used to examine consistency, and isolated acceptability judgements.

Table S3 provides the complete project-level evidence for RQ2 by comparing the exact MILP reference and the calibrated GA through solution quality, stochastic variability, optimum-hit counts, runtime, and speedup. Table S4 provides the complete project-level evidence for RQ3 by comparing the calibrated-GA recommendations with the historical teams through overall team fit, collaborative adequacy, and relative gain on the sixteen held-out 2025 projects.

Together, these tables support reproducibility and allow readers to trace the aggregate results and interpretations reported in the paper to their underlying project-level evidence.

Three additional tables give per-project detail that the main paper reports only in aggregate. Table S5 documents the adaptive-racing calibration of the GA configuration used throughout the paper (Section III-D). Table S6 provides the per-project best-of-run AE and AC for the four algorithmic baselines (GREEDY, RANDOM, RANDOM-SEARCH, and $GA_{UNIFORM}$) that Table II of the main paper reports only as sixteen-project means. Table S7 provides the same per-project detail for the selected direct-LLM response compared with the calibrated GA. None of these values were recalculated for this document; they are reproduced from the underlying result workbooks and their source-file manifests.

TABLE S1: Summary of the industrial dataset.

Aspect	Summary
Candidate pool	512 developers with at least one classified technical competence.
Technical taxonomy	13 domains, 213 ecosystems, and 23 programming languages. Median per developer: 2 domains, 6 ecosystems, and 3 languages.
Calibration projects	205 projects starting up to 2024. Team sizes: k=4: 45, k=5: 38, k=6: 37, k=7: 30, k=8: 26, k=9: 29.
Held-out projects	16 projects starting in 2025. Team sizes: k=4: 4, k=5: 4, k=6: 1, k=7: 1, k=8: 2, k=9: 4.
Collaboration graph	Pairwise collaboration history extracted from prior project co-participation and transformed through the expert-calibrated compatibility function.

TABLE S2: Retained practitioner-validation design.

Task	Unique cases	Repeats	Rows
Calibrated GA vs. historical	16	3	19
Calibrated GA vs. uniform GA	12	2	14
Calibrated-GA acceptability	16	–	16

TABLE S3: RQ2 comparison between the exact MILP reference and the calibrated GA on the twelve size-four controlled benchmark projects.

Project	MILP AE	GA best AE	Best gap (%)	GA AE mean ± SD	Hits/30	MILP time (s)	Speedup
P1	0.818783	0.818783	0.00	0.788474 ± 0.050642	12	109.69	7.85×
P2	0.844445	0.844445	0.00	0.769284 ± 0.059689	7	208.44	10.26×
P3	0.813492	0.810847	0.33	0.789697 ± 0.032128	0	186.75	7.61×
P4	0.864286	0.864286	0.00	0.817566 ± 0.033847	1	45.74	4.41×
P5	0.843915	0.826720	2.04	0.771497 ± 0.032369	0	34.46	2.20×
P6	0.781746	0.781746	0.00	0.753527 ± 0.035620	14	508.73	93.51×
P7	0.870900	0.859326	1.33	0.858016 ± 0.004189	0	585.89	117.37×
P8	0.857672	0.857672	0.00	0.790728 ± 0.054681	10	195.47	27.33×
P9	0.867725	0.867725	0.00	0.854559 ± 0.023670	16	122.19	15.06×
P10	0.900794	0.900794	0.00	0.876323 ± 0.024483	11	105.33	26.60×
P11	0.834656	0.834656	0.00	0.779374 ± 0.031629	2	73.92	12.84×
P12	0.877513	0.857672	2.26	0.794899 ± 0.028884	0	226.69	45.79×
Summary	–	–	0.50 mean	–	73/360	–	13.95× median

TABLE S4: RQ3 backtest on the sixteen held-out 2025 projects.

Project	k	$\mathrm{AE}_{\mathrm{real}}$	$\mathrm{AE}_{\mathrm{GA}}$	Gain (%)	$\mathrm{AC}_{\mathrm{real}}$	$\mathrm{AC}_{\mathrm{GA}}$
P1	9	0.5238	0.6114	16.72	0.5143	0.8111
P2	8	0.3750	0.5167	37.78	0.3643	0.6000
P3	5	0.4532	0.6822	50.53	0.4200	0.7200
P4	9	0.4008	0.6506	62.32	0.3571	0.8167
P5	4	0.5762	0.6365	10.47	0.6000	0.8667
P6	7	0.5246	0.5393	2.81	0.7667	0.7952
P7	5	0.5817	0.6326	8.75	0.6333	0.8600
P8	5	0.4365	0.6757	54.80	0.4000	0.8400
P9	6	0.4048	0.5311	31.20	0.4000	0.6467
P10	8	0.5869	0.6162	5.00	0.6643	0.8429
P11	9	0.3274	0.5041	53.98	0.3071	0.6833
P12	4	0.3770	0.5178	37.35	0.3667	0.7667
P13	4	0.7817	0.8643	10.56	0.7667	0.9000
P14	4	0.4008	0.9008	124.75	0.3000	0.9000
P15	5	0.3770	0.5347	41.84	0.3667	0.7400
P16	9	0.4373	0.4963	13.48	0.4810	0.7667
Mean	–	0.4728	0.6194	35.15	0.4818	0.7847

Among the eight controlled benchmark projects where the GA reached the certified MILP optimum, four recovered the identical team up to member ordering (P1, P6, P10, P11), and four reached a different team with the same optimal AE, indicating alternative optima under the same objective (P2, P4, P8, P9).

TABLE S6: Per-project best-of-run AE and AC for the four algorithmic baselines on the sixteen held-out 2025 projects (aggregated in Table II of the main paper). GREEDY runs once per project; RANDOM, RANDOM-SEARCH, and $GA_{UNIFORM}$ report the best of 30 sampled/executed teams. AT is omitted for space. For RANDOM and RANDOM-SEARCH, Table II of the main paper reports mean AT/AC computed over all sampled runs rather than over each project's best-AE team as shown here, so the AC means below differ slightly from Table II for those two methods; the AE means and the GREEDY/$GA_{UNIFORM}$ figures match Table II exactly.

Project	GREEDY AE	GREEDY AC	RANDOM AE	RANDOM AC	RANDOM-SEARCH AE	RANDOM-SEARCH AC	$GA_{UNIFORM}$ AE	$GA_{UNIFORM}$ AC
P1	0.3869	0.3500	0.3908	0.3556	0.5026	0.4889	0.6114	0.8111
P2	0.3393	0.3214	0.3683	0.3571	0.4441	0.4429	0.4975	0.7786
P3	0.3865	0.3400	0.4190	0.3800	0.6178	0.6400	0.6794	0.8600
P4	0.3763	0.3278	0.3988	0.3556	0.5106	0.4889	0.6506	0.8167
P5	0.5119	0.5000	0.4563	0.4333	0.6225	0.6667	0.6429	0.9000
P6	0.3690	0.3476	0.3843	0.3667	0.4809	0.4810	0.5393	0.7952
P7	0.3952	0.3600	0.4119	0.3800	0.5935	0.7200	0.6165	0.8800
P8	0.4198	0.3800	0.4196	0.3800	0.6340	0.7200	0.6728	0.8600
P9	0.3881	0.3800	0.3992	0.3933	0.4628	0.5000	0.5268	0.7800
P10	0.5000	0.4857	0.3933	0.3571	0.5022	0.4929	0.6162	0.8429
P11	0.3446	0.3278	0.3677	0.3556	0.4386	0.4889	0.4923	0.8111
P12	0.3492	0.3333	0.4294	0.4333	0.4707	0.6667	0.5121	0.9000
P13	0.4206	0.3333	0.4778	0.4333	0.6905	0.6667	0.8643	0.9000
P14	0.4008	0.3000	0.5079	0.4333	0.7897	0.7667	0.9008	0.9000
P15	0.3214	0.3000	0.3891	0.3800	0.4937	0.5400	0.5255	0.9000
P16	0.3585	0.3444	0.3668	0.3556	0.4386	0.4889	0.4950	0.7278
Mean	0.3918	0.3582	0.4113	0.3844	0.5433	0.5787	0.6152	0.8415

TABLE S5: GA calibration via adaptive racing (Section III-D).

Aspect	Summary
Chronological split	205 pre-2025 projects: 144 training, 31 validation, 30 test.
Selected configuration	Population 150, 200 generations, elitism 3, mutation rate 0.005, crossover rate 1.0, 20 stable generations, CCC crossover operator.
Baseline configuration	Population 100, 100 generations, elitism 1, mutation rate 0.05, crossover rate 1.0, 10 stable generations, CCC crossover operator.
Test-set comparison	150 paired test-set runs per configuration. Selected mean fitness 0.6120; baseline mean fitness 0.5983; relative gain 2.29%; paired outcomes 108 wins, 35 losses, 7 ties.

TABLE S7: RQ5 blinded pairwise comparisons: AT/AC of the calibrated and $GA_{UNIFORM}$ teams for the five projects where the participant selected $GA_{UNIFORM}$ over the calibrated GA. In every case the uniform team had higher AC, lower AT, and a larger $|AT - AC|$ gap.

Project	Calibrated AT	Calibrated AC	Uniform AT	Uniform AC
P2	0.5000	0.6000	0.4413	0.7786
P7	0.5872	0.8600	0.5638	0.8800
P8	0.6429	0.8400	0.6354	0.8600
P11	0.4683	0.6833	0.4286	0.8111
P12	0.4680	0.7667	0.4345	0.9000

a) Direct-LLM input sanitization protocol.: The developer file provided to the direct-LLM baseline (Section RQ4 of the main paper) contained 510 anonymized developers, aligned with the candidate set present in the collaboration graph; two records without graph nodes were excluded. To prevent direct target-team leakage, 91 project-history links to the sixteen target project identifiers were removed from 78 developer records before the file was given to the LLM. The collaboration graph itself was otherwise preserved unchanged. The full sanitization manifest, listing every removed link by developer and project identifier, is preserved in the replication package.

TABLE S8: Per-project AT, AC, and AE for the selected direct-LLM response (response_1) and the calibrated-GA best-of-30 recommendation on the sixteen held-out 2025 projects (aggregated in Table II of the main paper). All sixteen LLM teams passed the validity checks (known developer identifiers, no duplicates, exact team size, full MUST coverage).

Project	LLM AT	LLM AC	LLM AE	GA best AT	GA best AC	GA best AE	AE gap (GA − LLM)
P1	0.5714	0.4167	0.4425	0.5714	0.8111	0.6114	0.1689
P2	0.4286	0.4000	0.4048	0.5000	0.6000	0.5167	0.1119
P3	0.6190	0.6200	0.6192	0.6746	0.7200	0.6822	0.0630
P4	0.6190	0.4389	0.4689	0.6174	0.8167	0.6506	0.1817
P5	0.5714	0.7000	0.5929	0.5905	0.8667	0.6365	0.0437
P6	0.4762	0.5381	0.4865	0.4881	0.7952	0.5393	0.0528
P7	0.5714	0.5000	0.5119	0.5872	0.8600	0.6326	0.1207
P8	0.6190	0.5200	0.5365	0.6429	0.8400	0.6757	0.1392
P9	0.4286	0.5533	0.4494	0.5079	0.6467	0.5311	0.0817
P10	0.5714	0.5214	0.5298	0.5709	0.8429	0.6162	0.0864
P11	0.4286	0.4611	0.4340	0.4683	0.6833	0.5041	0.0701
P12	0.4286	0.5667	0.4516	0.4680	0.7667	0.5178	0.0662
P13	0.6905	0.5000	0.5317	0.8571	0.9000	0.8643	0.3326
P14	0.9048	0.5667	0.6230	0.9048	0.9000	0.9008	0.2778
P15	0.4286	0.4000	0.4048	0.4937	0.7400	0.5347	0.1299
P16	0.4286	0.4833	0.4377	0.4422	0.7667	0.4963	0.0586
Mean	0.5491	0.5116	0.4953	0.5866	0.7847	0.6194	0.1241